

DCL (Data Control Language) Commands in Oracle

DCL commands are used to control access to database objects and manage user privileges. In Oracle, the primary DCL commands are:

- GRANT
- REVOKE

These commands help enforce database security and authorization.

1. GRANT Command

The GRANT command is used to assign privileges to users or roles.

Privileges are of two types:

1. System Privileges
2. Object Privileges

1.1 Granting System Privileges

System privileges allow a user to perform specific database-wide actions.

Syntax

GRANT system_privilege TO user_or_role;

Example 1: Single System Privilege

GRANT CREATE TABLE TO RAHUL;

RAHUL can now create tables in his own schema.

Example 2: Multiple System Privileges

GRANT CREATE TABLE, CREATE VIEW, CREATE SEQUENCE TO RAHUL;

Example 3: Granting to Multiple Users

GRANT CREATE TABLE TO RAHUL, AMIT, SNEHA;

Example 4: Grant with ADMIN OPTION

GRANT CREATE TABLE TO RAHUL WITH ADMIN OPTION;

RAHUL can now grant CREATE TABLE to other users.

Example 5: Granting ALL PRIVILEGES (System Level)

GRANT ALL PRIVILEGES TO RAHUL;

This grants all system privileges allowed to the current user.

Common System Privileges in Oracle

- CREATE SESSION
- CREATE TABLE
- CREATE VIEW
- CREATE USER
- CREATE PROCEDURE
- ALTER USER
- DROP USER
- CREATE ROLE
- UNLIMITED TABLESPACE

Example:

```
GRANT CREATE SESSION TO RAHUL;
```

A user must have CREATE SESSION to log in.

1.2 Granting Object Privileges

Object privileges allow actions on specific objects like tables, views, sequences, procedures.

A. Granting Privileges on Tables

Syntax

```
GRANT privilege ON object_name TO user;
```

Example 1: Grant SELECT

```
GRANT SELECT ON EMP TO RAHUL;
```

Example 2: Grant Multiple Privileges

```
GRANT SELECT, INSERT, UPDATE ON EMP TO RAHUL;
```

Example 3: Grant UPDATE on Specific Columns

```
GRANT UPDATE (SALARY, COMMISSION) ON EMP TO RAHUL;
```

Example 4: Grant with GRANT OPTION

```
GRANT SELECT ON EMP TO RAHUL WITH GRANT OPTION;
```

RAHUL can now grant SELECT on EMP to others.

Example 5: Grant to Multiple Users

```
GRANT SELECT ON EMP TO RAHUL, AMIT;
```

Example 6: Grant to PUBLIC

```
GRANT SELECT ON EMP TO PUBLIC;
```

All users can access EMP.

B. Granting Privileges on Views

```
GRANT SELECT ON EMP_VIEW TO RAHUL;
```

C. Granting EXECUTE on Procedures/Functions

```
GRANT EXECUTE ON CALCULATE_SALARY TO RAHUL;
```

D. Granting Privileges on Sequences

```
GRANT SELECT ON EMP_SEQ TO RAHUL;
```

1.3 Granting Roles

Roles are named groups of privileges.

Create Role

```
CREATE ROLE manager_role;
```

Grant Privileges to Role

GRANT SELECT, INSERT ON EMP TO manager_role;

Grant Role to User

GRANT manager_role TO RAHUL;

Grant Role with ADMIN OPTION

GRANT manager_role TO RAHUL WITH ADMIN OPTION;

Set Default Role

ALTER USER RAHUL DEFAULT ROLE manager_role;

2. REVOKE Command

The REVOKE command removes privileges from users or roles.

2.1 Revoke System Privileges

Syntax

REVOKE system_privilege FROM user_or_role;

Example 1

REVOKE CREATE TABLE FROM RAHUL;

Example 2: Revoke Multiple Privileges

REVOKE CREATE TABLE, CREATE VIEW FROM RAHUL;

Example 3: Revoke ALL PRIVILEGES

REVOKE ALL PRIVILEGES FROM RAHUL;

2.2 Revoke Object Privileges

Syntax

REVOKE privilege ON object_name FROM user;

Example 1

REVOKE SELECT ON EMP FROM RAHUL;

Example 2: Revoke Multiple Privileges

REVOKE SELECT, INSERT ON EMP FROM RAHUL;

Example 3: Revoke Column-Level Privilege

REVOKE UPDATE (SALARY) ON EMP FROM RAHUL;

Example 4: Revoke Role from User

REVOKE manager_role FROM RAHUL;

Cascading Revocation

If a user was granted a privilege WITH GRANT OPTION and granted it to others, revoking it from that user also removes it from the users to whom it was granted.

Example:

```
GRANT SELECT ON EMP TO RAHUL WITH GRANT OPTION;  
GRANT SELECT ON EMP TO AMIT;  
REVOKE SELECT ON EMP FROM RAHUL;  
AMIT also loses the SELECT privilege.
```

Granting and Revoking from PUBLIC

```
GRANT SELECT ON EMP TO PUBLIC;  
REVOKE SELECT ON EMP FROM PUBLIC;
```

Checking Granted Privileges

Check User System Privileges

```
SELECT * FROM USER_SYS_PRIVS;
```

Check Object Privileges

```
SELECT * FROM USER_TAB_PRIVS;
```

Check Role Privileges

```
SELECT * FROM USER_ROLE_PRIVS;
```

Important Points in Oracle

1. DCL statements are auto-commit.
2. Only privileged users (like DBA) can grant system privileges.
3. WITH ADMIN OPTION applies to system privileges and roles.
4. WITH GRANT OPTION applies to object privileges.
5. Privileges can be granted to users, roles, and PUBLIC.
6. Column-level privileges are allowed only for UPDATE and INSERT.
7. You cannot grant privileges on objects you do not own unless you have GRANT OPTION.

Complete Example Scenario

Step 1: Create User

```
CREATE USER RAHUL IDENTIFIED BY rahul123;
```

Step 2: Allow Login

```
GRANT CREATE SESSION TO RAHUL;
```

Step 3: Grant Table Access

```
GRANT SELECT, INSERT ON EMP TO RAHUL;
```

Step 4: Revoke Insert

```
REVOKE INSERT ON EMP FROM RAHUL;
```

Step 5: Remove Login Access

```
REVOKE CREATE SESSION FROM RAHUL;
```